

1. Complete the following table

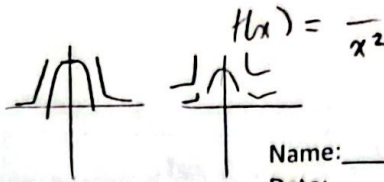
[11]

Function	a) $f(x) = \frac{4}{-x+2}$	b) $g(x) = \frac{16}{x^2+2x-8}$	c) $h(x) = \frac{2x-4}{x-1}$
Horizontal Asymptote(s)	$y = 0$	$y = 0$	$y = \frac{2}{1} = 2$
Vertical Asymptote(s)	$-x+2=0$ $x=2$	$(x-2)(x+4)$ $x=2, -4$	$x=1$
y-intercept	$f(x) = \frac{4}{0+2}$ $= 2$ $(0, 2)$	$f(x) = \frac{16}{(0)^2+2(0)-8}$ $= -2$ $(0, -2)$	$f(x) = \frac{2(0)-4}{0-1}$ $= 4$ $(0, 4)$
x-intercept	N/A	N/A	$2x-4=0$ $2x=4$ $x=2$ (2, 0)
Domain	$\{x \in \mathbb{R} \mid x \neq 2\}$	$\{x \in \mathbb{R} \mid x \neq -4, 2\}$	$\{x \in \mathbb{R} \mid x \neq 1\}$
Range	$\{y \in \mathbb{R} \mid y \neq 0\}$	$\{y \in \mathbb{R} \mid y \neq 0, y > 0, y \leq -\frac{16}{9}\}$	$\{y \in \mathbb{R} \mid y \neq 2\}$
Sketch Label: -asymptotes -intercepts			

$$\begin{array}{ll}
 2^+ & -\infty \\
 2^- & +\infty \\
 +\infty & 0^- \\
 -\infty & 0^+
 \end{array}$$

$$\begin{array}{lll}
 2^+ & +\infty & 1^+ & 0^+ \\
 2^- & -\infty & -\infty & 0^+ \\
 -4^+ & -\infty & & \\
 -4^- & +\infty & &
 \end{array}$$

$$\begin{array}{ll}
 1^+ & -\infty \\
 1^- & +\infty \\
 +\infty & 2^- \\
 -\infty & 2^+
 \end{array}$$



Name: _____

Date: _____

2. For the function $g(x) = \frac{16}{x^2 + 2x - 8}$ (question 1b), determine the following intervals. [3]

a)

$f(x) > 0$	$x < -4$, $x > 2$ incorrect notation
$f(x) < 0$	$-4 < x < 2$

b)

positive slope	$x < -4$, $-4 < x \leq -1$
negative slope	$-1 \leq x < 2$, $x > 2$

3. A function is described as the reciprocal of a quadratic function with the following characteristics. Determine a possible equation of the function. [3]

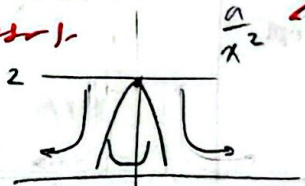
- Domain: $\{x \in \mathbb{R} | x \neq -3\}$
- Range: $\{y \in \mathbb{R} | 0 < y < 2\}$
- Y-intercept: $(0, 2)$ vertex

no VA: $\frac{a+b}{x^2}$

H/A: $2 = \frac{2x^2 + b}{x^2}$

$2 = \frac{0x^2 + b}{0x^2}$

$f(x) = \frac{2x^2 + 2}{x^2}$



$f(x) = \frac{8}{x^2 + 4}$

no VA

4. Determine the equation for the following function in the form $f(x) = \frac{ax+b}{cx+d}$ [3]

$\frac{ax+b}{cx+d}$

VA: $-1 \rightarrow x+1$ $-\frac{d}{c} \rightarrow -\frac{1}{1} = -1$

H/A: $\frac{2}{1} = 2$ $\frac{a}{c} \rightarrow \frac{2}{1} = 2$

$\frac{2x+b}{x+1}$

y-int: $-3 = \frac{2(0)+b}{0+1}$

$f(x) = \frac{2x-3}{x+1}$

